MIL-C-17/171A
18 July 1985
SUPERSEDING
MIL-C-17/171
23 February 1982

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/171-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-17.

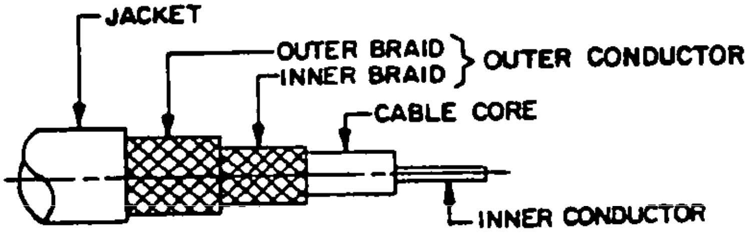

FIGURE 1. Configuration.

TABLE I. Description.

Component	Construction details
Inner conductor	Silver covered, copper-covered steel wire. Overall diameter: .059 inch ±.001.
Ⓐ Cable core	Type F-1: Solid, extruded PTFE Diameter: .185 inch ±.005.
Outer conductor	Double braid of AWG No. 34 silver-covered wire. Diameter: .250 inch maximum.
Inner braid	Carriers : 24 Ends : 5 Picks/inch: 14.5 ±10%
Outer braid	Carriers : 24 Ends : 6 Picks/inch: 11.5 ±10%
Jacket	Type IX: FEP. Overall diameter: .280 inch ±.008. Minimum wall thickness: .015 inch.

CAUTION IS DIRECTED TO THE APPLICATION OF THIS CABLE ABOVE 400 MHZ. ATTENUATION IS TESTED ONLY AT 400 MHZ. SRL AND POWER HANDLING CAPABILITIES ARE NOT STIPULATED HEREIN.

Ⓐ denotes changes

FSC 6145

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

ENGINEERING INFORMATION:

Continuous working voltage: 2,200 V rms, maximum.

Velocity of propagation: 69.5 percent, nominal.

Operating temperature range: -55°C to +200°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.9 ohm per 100 feet.

Elongation: 1 percent, minimum.

Tensile strength: 110 $klb_f/inch^2$, minimum.

Engineering note: This cable is useful in general purpose, high temperature applications. (See connector series "TNC" and "BNC" per MIL-C-39012.)

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: Not applicable.

Adhesion of conductors:

Inner conductor to core: 4 pounds, minimum; 30 pounds, maximum.

Aging stability: Not applicable.

Stress crack resistance: +230°C ±5°C; mandrel size seven times the jacket diameter.

Outer conductor integrity: Not applicable.

Ⓐ Cold bend: -55°C ±2°C.

Dimensional stability: +200°C ±5°C.

Inner conductor from core: 0.250 inch, maximum.

Inner conductor from jacket: 0.312 inch, maximum.

Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Applicable.

Weight: 0.0924 pound per foot, maximum.

Electrical:

Continuity: Applicable.

Ⓐ Spark test: 2,000 V rms, +10%, -0%.

Ⓐ Voltage withstanding: 5,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 3,000 V rms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: 6.4 dB/100 ft maximum at 400 MHz.

Structural return loss: Not applicable.

Capacitance: 32 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross-reference of part number.

Part number	Superseded part number or type designation
M17/171-00001	RG-304 per MIL-C-17D

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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